

# Excel Advanced Analytics Workbook: 10 Expert Challenges

Target Framework: Complex Array-Style Logic and Dynamic Matrix Evaluation (Q51-Q60)

**Capstone Assignment Notice:** This final tier introduces dynamic multi-condition string lookups, running percentage allocations, conditional counting across ranges, and compound text mining techniques. Use these to stress-test your students' computational understanding of Excel.

## Advanced Analytics Exercises (Q51–Q60)

### Q51 Dynamic Conditional Duplication Auditing

In column L, write an advanced rolling analytical formula to track duplicate transactions. For every row, if the Order\_ID has appeared previously above it, display **"Duplicate Flag"**. If it is the first time the ID appears, display **"Unique"**.

```
=IF(COUNTIF($A$2:A2, A2)>1, "Duplicate Flag", "Unique")
```

Hint: Notice the expanding range locking technique (\$A\$2:A2) which naturally extends downwards as the formula is filled.

### Q52 Extracting Revenue Contribution Share

In column M (headed **Revenue\_Share\_%**), write an equation that calculates what percentage of the total corporate earnings (\$250 orders combined) this single line item represents. Format the output style as a Percentage with 2 decimal places.

```
=I2 / SUM($I$2:$I$251)
```

### Q53 Extracting Variable Domain Text via Index Mining

Assume an order's payment processing server address string is formatted inside your notes as "User\_ID@Domain.com". If column H contains email-like sequences, use string text pointers to extract strictly the domain text trailing *after* the "@" character marker.

```
=MID(H2, FIND("@", H2) + 1, LEN(H2) - FIND("@", H2))
```

### Q54 Multi-Tiered Conditional Pricing Adjustments

Write an advanced logical expression to apply localized tariffs on total revenue based on compound geography: If the region is **North** AND the product is an **Air Fryer**, apply a 12% surcharge; if the region is **South** OR payment method is **Apple Pay**, deduct a 4% discount; otherwise, leave the base total revenue unchanged.

```
=IF(AND(C2="North", E2="Air Fryer"), I2*1.12, IF(OR(C2="South", H2="Apple Pay"), I2*0.96, I2))
```

### Q55 Locating the Highest Order Code via Dynamic Matching

Construct a compound formula that automatically combs through your financial matrix, identifies the absolute highest single Total\_Revenue invoice amount, and returns the string text of its associated Order\_ID.

```
=INDEX(A2:A251, MATCH(MAX(I2:I251), I2:I251, 0))
```

#### Q56 Extracting First Names from Variable Full-String Entries

If a customer column displays values as "Firstname Lastname" separated by a single blank space, write a text manipulation formula to extract only the isolated first word string block dynamically.

```
=LEFT(E2, FIND(" ", E2) - 1)
```

#### Q57 Multi-Criteria Dynamic Filter Extraction (FILTER)

Using modern Excel dynamic array operations, extract a mini-table to an external summary sheet displaying all columns, but filtered down to only show records where the region is **East** AND the Quantity is greater than or equal to 4.

```
=FILTER(A2:H251, (C2:C251="East") * (F2:F251>=4))
```

*Hint: The asterisk symbol (\*) acts as the mathematical logical "AND" operator inside dynamic array functions.*

#### Q58 Advanced Multi-Condition Lookup Matching (XLOOKUP Array Intersects)

Locate an active invoice item based on multiple independent criteria. Look up a transaction that simultaneously matches **\*\*Region = "South" AND \*\*Product = "Backpack"\*\*, and extract its matching unique `Order\_ID`.**

```
=XLOOKUP(1, (C2:C251="South") * (E2:E251="Backpack"), A2:A251)
```

#### Q59 Eliminating Zero-Quantity Volatility via Blank Blanking

Write a protective nested expression: evaluate if a transaction row's Quantity cell evaluates to empty or zero. If true, display an entirely blank string cell (" ") to keep clean visual aesthetics; if false, execute the full revenue multiplication matrix formula.

```
=IF(OR(ISBLANK(F2), F2=0), "", F2 * G2)
```

#### Q60 Evaluating Aggregate Average Over Median Constraints

Write a compound array calculation to find the sum total of all Total\_Revenue cell lines, but strictly targeting and summing rows where the individual order revenue is greater than the overall system-wide Median value.

```
=SUMIF(I2:I251, ">" & MEDIAN(I2:I251))
```